Supplementary material: Baryonic Suppression of the $z = 0$ Matter Power Spectrum: HACC Emulator Predictions and Weak-Lensing Context

HEP-Genesis-Agent session record
Project: ramp-combo-baryonic-pk

2026-10-23

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `3b089828`
- Started: 2026-10-17 14:31:53; report generated: 2026-10-23 14:42:03
- Project: ramp-combo-baryonic-pk (`/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk`)
- Compute execution site(s) selected in the session: local
- 2 user turn(s), 192 tool call(s) (0 returned errors), 3 figure(s), 27 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
report	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`serve-sbbn.sh list_sterile_nu_skills`	`{}`	`{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...`
2	`serve-hydro.sh list_observables`	`{}`	`{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...`
3	`hep-research-aid list_agent_files`	`{}`	`[]`
4	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
5	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
6	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - workspace`
7	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "baryonic suppression matter power spectrum emulator AGN feedback", "max_results": 8}	[0411485v1] Obscured AGN in Multiwavelength Surveys Abstract: Using a simple unification model for AGN, with a fixed ratio of obscured to unobscured AGN of 3: 1, we explain the X-ray, optical and inf...
9	local search_arxiv_ abstracts	{"query": "baryonic feedback weak lensing S8 cosmic shear power spectrum suppression", "max_results": 8}	[0409198v2] Effect of Hot Baryons on the Weak-Lensing Shear Power Spectrum Abstract: We investigate the impact of the intracluster medium on the weak-lensing shear power spectrum (PS). Using a halo ...
10	local search_arxiv_ abstracts	{"query": "baryon fraction in clusters predicts power spectrum suppression", "max_ results": 6}	[0105520v1] The baryon fraction distribution in X-rays galaxy clusters Abstract: The baryon fraction in galaxy clusters is one of the most direct way to constrain $\omo$. The baryonic fraction is es...
11	local search_arxiv_ abstracts	{"max_results": 6, "query": "FLAMINGO simulations baryonic effects matter power spectrum S8 tension"}	[2410.17109v2] The FLAMINGO project: Baryon effects on the matter power spectrum Abstract: The effect of baryon physics associated with galaxy formation onto the large-scale matter distribution of the...
12	local search_arxiv_ abstracts	{"max_results": 6, "query": "BACCO emulator baryonic effects cosmic shear DES KiDS constraints"}	[2305.17173v2] DES Y3 + KiDS-1000: Consistent cosmology combining cosmic shear surveys Abstract: We present a joint cosmic shear analysis of the Dark Energy Survey (DES Y3) and the Kilo-Degree Survey ...

#	Tool	Arguments	Outcome
13	local search_arxiv_ abstracts	{"max_results": 6, "query": "kinetic Sunyaev-Zeldovich strong feedback suppression cosmic shear"}	[0206245v1] Cosmic Shear: The Dark Side of the Universe Abstract: We discuss the present status and future prospects for cosmic shear observations and their cosmological constraints. We review the e...
14	local search_arxiv_ abstracts	{"max_results": 5, "query": "Amon Efstathiou non-linear solution S8 tension baryonic feedback"}	[2305.09827v1] A non-linear solution to the S_8 tension II: Analysis of DES Year 3 cosmic shear Abstract: Weak galaxy lensing surveys have consistently reported low values of the S_8 parameter com...
15	local search_arxiv_ abstracts	{"max_results": 5, "query": "van Daalen effects of galaxy formation on the matter power spectrum OWLS"}	[1104.1174v2] The effects of galaxy formation on the matter power spectrum: A challenge for precision cosmology Abstract: Upcoming weak lensing surveys, such as LSST, EUCLID, and WFIRST, aim to measur...
16	local get_arxiv_ metadata	{"arxiv_id": "2410.17109"}	arXiv ID: 2410.17109 Title: The FLAMINGO project: Baryon effects on the matter power spectrum Authors: Matthieu Schaller, Joop Schaye, Roi Kugel, Jeger C. Broxterman, Marcel P. van Daalen Year: 2024 A...
17	local search_arxiv_ abstracts	{"max_results": 5, "query": "SP(k) baryon fraction halo mass power spectrum suppression relation Salcido"}	[2604.05701v1] Measurement of the CKM angle γ in $B^{\pm} \ rightarrow D(\rightarrow K^{0}_{\rm S} h^{\prime+} h^{\prime-})h^{\pm}$ decays with a novel approach Abstract: A measurement of the CKM angl...

#	Tool	Arguments	Outcome
18	local search_arxiv_ abstracts	{"max_results": 5, "query": "weak lensing kSZ joint constraint baryonic feedback stronger than simulations"}	[2406.08919v3] Strong gravitational lenses from the Vera C. Rubin Observatory Abstract: Like many areas of astrophysics and cosmology, the Vera C. Rubin Observatory will be transformational for almost...
19	local get_arxiv_ metadata	{"arxiv_id": "2512.02954"}	arXiv ID: 2512.02954 Title: The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy lensing Authors: Jared Siegel, Leah Bigw...
20	serve-hydro.sh describe_ parameters	{"stat_name": "Pk-ratio"}	{ "status": "success", "message": "7-parameter design space for Pk-ratio: kappa_w, e_w, M_seed, v_kin, epsilon_ kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { "ka...
21	serve-hydro.sh list_observables	{}	{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...
22	local download_arxiv_ full_text	{"arxiv_id": "2309.07959", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/runs/ 261017_143153"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261017_143153/ 2309.07959.txt

#	Tool	Arguments	Outcome
23	local download_arxiv_full_text	{"arxiv_id": "2410.17109", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt
24	serve-hydro.sh predict_observable	{"stat_name": "Pk-ratio", "omega_m": 0.14, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-bar...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducia...
25	serve-hydro.sh predict_observable	{"stat_name": "Pk-ratio", "omega_m": 0.14, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 1.0, "m_seed": 1.3, "v_kin": 1.2, "eps_kin": 1.2, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryo...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/strongA...
26	serve-hydro.sh predict_observable	{"stat_name": "Pk-ratio", "omega_m": 0.14, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.2, "m_seed": 1.3, "v_kin": 0.1, "eps_kin": 0.02, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-bary...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/weakAGN...
27	local download_arxiv_full_text	{"arxiv_id": "2206.11794", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2206.11794.txt

#	Tool	Arguments	Outcome
28	hep-research-aid load_table_data	{"filename": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducial_z0/Pk-ratio_z0.0000_prediction.csv"}	Error: could not convert string 'x' to float64 at row 0, column 1.
29	local download_arxiv_full_text	{"arxiv_id": "2512.02954", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2512.02954.txt
30	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducial_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_l...
31	local download_arxiv_full_text	{"arxiv_id": "2512.02954", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2512.02954.txt
32	local download_arxiv_full_text	{"arxiv_id": "2309.07959", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.07959.txt
33	serve-hydro.sh plot_observable_comparison	{"prediction_files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducial_z0/Pk-ratio_z0.0000_prediction.csv", "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo...	{ "status": "success", "message": "Plotted 3 Pk-ratio predictions.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/Pk-ratio_compariso... [Fig. 8]

#	Tool	Arguments	Outcome
34	serve-hydro.sh plot_prediction	{"e_w": 0.6, "eps_kin": 0.61, "kappa_w": 3, "m_seed": 1.3, "omega_m": 0.14, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153", "sigma_8": 0.8, "stat_name": "Pk-rat...	{ "status": "success", "message": "Plotted Pk-ratio prediction at z=0.0000.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/Pk-ratio_... [Fig. 9]
35	local download_arxiv_full_text	{"arxiv_id": "2206.11794", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2206.11794.txt
36	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.07959.txt", "length": 4000}	[chars 0-4000 of 141592] (more available, next start=4000) Title: The FLAMINGO project: revisiting the S_8 tension and the role of baryonic physics Authors: Ian G. McCarthy, Jaime Salcido, Joop Sch...
37	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/strongAGN_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=1.0, m_seed=1.3, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_lab...
38	local download_arxiv_full_text	{"arxiv_id": "2309.02920", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.02920.txt

#	Tool	Arguments	Outcome
39	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2206.11794.txt", "length": 4000}	[chars 0-4000 of 65221] (more available, next start=4000) Title: A non-linear solution to the S_8 tension? Authors: Alexandra Amon, George Efstathiou Year: 2022 arXiv ID: 2206.11794 ==========...
40	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2512.02954.txt", "length": 4000}	[chars 0-4000 of 100641] (more available, next start=4000) Title: The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy l...
41	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2206.11794.txt", "start": 44000, "length": 9000}	[chars 44000-53000 of 65221] (more available, next start=53000) ross-correlations between cosmic shear and the kinetic and thermal Sunyaev-Zeldovich (SZ) e?ects (e.g. Schneider et al. 2021; Tr¨ oster...
42	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2512.02954.txt", "start": 62000, "length": 8000}	[chars 62000-70000 of 100641] (more available, next start=70000) andhalomodelframeworksarelargelyconsistent. The?(?)/? DM Only(?)posterior from La Posta et al. (2025) marginally deviates from the o...
43	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt", "length": 4000}	[chars 0-4000 of 85813] (more available, next start=4000) Title: The FLAMINGO project: Baryon effects on the matter power spectrum Authors: Matthieu Schaller, Joop Schaye, Roi Kugel, Jeger C. Broxter...

#	Tool	Arguments	Outcome
44	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2512.02954.txt", "length": 4000}	[chars 0-4000 of 100641] (more available, next start=4000) Title: The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy l...
45	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.07959.txt", "length": 4000}	[chars 0-4000 of 141592] (more available, next start=4000) Title: The FLAMINGO project: revisiting the S_8 tension and the role of baryonic physics Authors: Ian G. McCarthy, Jaime Salcido, Joop Sch...
46	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2206.11794.txt", "length": 4000}	[chars 0-4000 of 65221] (more available, next start=4000) Title: A non-linear solution to the S_8 tension? Authors: Alexandra Amon, George Efstathiou Year: 2022 arXiv ID: 2206.11794 ==========...
47	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.02920.txt", "length": 4000}	[chars 0-4000 of 76364] (more available, next start=4000) Title: Determining the Baryon Impact on the Matter Power Spectrum with Galaxy Clusters Authors: Sebastian Grandis, Giovanni Arico', Aurel Sch...
48	board post_finding	{"claim": "HACC emulator Pk-ratio (P_hydro/P_grav) at z=0, cosmology fixed (omega_m=0.14, sigma_8=0.8, kappa_w=3, M_seed=1.3). Fiducial (e_w=0.6, v_kin=0.65, eps_kin=0.61): 0.99968 at k=0.101, 0.99197 at k=0.510, 0.97832...	Recorded finding f-001.

#	Tool	Arguments	Outcome
49	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/weakAGN_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=0.2, m_seed=1.3, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_la...
50	local cite	{"arxiv_id": "2309.07959", "note": "[\"non-linear scales, but that it is unlikely that baryon physics is driving this modification.\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-p...	Cited: The FLAMINGO project: revisiting the S_8 tension and the role of baryonic physics (\cite{arxiv230907959})
51	local cite	{"arxiv_id": "2206.11794", "note": "[\"with a value Amod ≈ 0.69\", \"is required if the Planck ΛCDM\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/references....	Cited: A non-linear solution to the S_8 tension? (\cite{arxiv220611794})
52	local cite	{"arxiv_id": "2512.02954", "note": "[\"is significantly biased without a baryon model.\", \"and eROSITA calibrated baryonification prior (blue) recovers half the preci\"]", "bib_path": "/Users/nramachandra/hep-genesis-te...	Cited: The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy lensing (\cite{arxiv251202954})
53	board post_finding	{"claim": "The HACC Pk-ratio at z=0 for the weak-AGN point (kappa_w=3, e_w=0.2, M_seed=1.3, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) shows no suppression. Instead it shows an enhancement: 1.00009 at k=0.101, 1...	Recorded finding f-002.

#	Tool	Arguments	Outcome
54	board post_finding	{"claim": "Literature synthesis from full texts that were read: (1) Mechanism: energy from active galactic nuclei (AGN) moves gas relative to the dark matter. This suppresses the nonlinear matter power by tens of percent...	Recorded finding f-003.
55	local cite	{"arxiv_id": "2410.17109", "note": "[\"We find that lower gas fractions in groups and clusters lead to a stronger response and that the use of\", \"We find only a small (one per cent at k< 10 h· Mpc-1)\"]", "bib_path": "...	Cited: The FLAMINGO project: Baryon effects on the matter power spectrum (\ cite{arxiv241017109})
56	local cite	{"arxiv_id": "2309.07959", "note": "[\"non-linear scales, but that it is unlikely that baryon physics is driving this modification.\ "]", "bib_path": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-p...	Cited: The FLAMINGO project: revisiting the S_8 tension and the role of baryonic physics (\cite{arxiv230907959})
57	local cite	{"arxiv_id": "2206.11794", "note": "[\"argue that these observations can be reconciled with the Planck ΛCDM cosmology\ "]", "bib_path": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/runs/ 261017_...	Cited: A non-linear solution to the S_8 tension? (\ cite{arxiv220611794})

#	Tool	Arguments	Outcome
58	board post_finding	{"claim": "Literature (quoted, cited): Amon & Efstathiou (2206.11794) find KiDS-1000 xi_pm is consistent with Planck LCDM only if non-linear power is suppressed with A_mod ~ 0.69, extending to k ~ 0.2 h/Mpc, outside the ...	Recorded finding f-004.
59	board post_finding	{"claim": "This is a qualitative comparison only, and the d-01 HACC emulator values were not on the board when it was posted. Any emulator suppression at k=1 h/Mpc should be read against these benchmarks: about 5% (HSC-X...	Recorded finding f-005.
60	board post_question	{"text": "Still open from d-02: four items were not covered. (1) The exact depth and k of FLAMINGO's z=0 suppression minimum, from the body or figures of 2410.17109. (2) Extraction of the 2410.17109 figure comparing feed...	Recorded question q-05.
61	board post_question	{"text": "The Pk-ratio CSVs in runs/261017_143153/{fiducial,strongAGN,weakAGN}_z0 have 799 k-bins, but the text reader truncates near k=3.7 h/Mpc and load_table_data rejects the header row. Can someone with a proper CSV ...	Recorded question q-06.
62	serve-hydro.sh describe_parameters	{"stat_name": "Pk-ratio"}	{ "status": "success", "message": "7-parameter design space for Pk-ratio: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { "ka...

#	Tool	Arguments	Outcome
63	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.02920.txt", "start": 0, "length": 3000}	[chars 0-3000 of 76364] (more available, next start=3000) Title: Determining the Baryon Impact on the Matter Power Spectrum with Galaxy Clusters Authors: Sebastian Grandis, Giovanni Arico', Aurel Sch...
64	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt", "start": 0, "length": 3000}	[chars 0-3000 of 85813] (more available, next start=3000) Title: The FLAMINGO project: Baryon effects on the matter power spectrum Authors: Matthieu Schaller, Joop Schaye, Roi Kugel, Jeger C. Broxter...
65	local download_arxiv_full_text	{"arxiv_id": "2305.09827", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2305.09827.txt
66	serve-hydro.sh predict_observable	{"stat_name": "Pk-ratio", "omega_m": 0.14, "sigma_8": 0.8, "kappa_w": 2.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-bar...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw2...
67	serve-hydro.sh predict_observable	{"stat_name": "Pk-ratio", "omega_m": 0.14, "sigma_8": 0.8, "kappa_w": 4.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-bar...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw4...

#	Tool	Arguments	Outcome
68	local download_arxiv_full_text	{"arxiv_id": "2410.12500", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.12500.txt
69	local cite	{"arxiv_id": "2309.02920", "note": "[\"In our fiducial setting, we also predict at 97.5 percent credibility, that at scales\", \"baryon feedback impacts the matter power less than\", \"This puts into question if baryon f...	Cited: Determining the Baryon Impact on the Matter Power Spectrum with Galaxy Clusters (\cite{arxiv230902920})
70	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt", "start": 30000, "length": 12000}	[chars 30000-42000 of 85813] (more available, next start=42000) r can be used in cosmology model inference searches (e.g. using an MCMC sampler) without leading to problematic time overheads. It coul...
71	serve-hydro.sh predict_observable	{"stat_name": "Pk-ratio", "omega_m": 0.14, "sigma_8": 0.8, "kappa_w": 3.0, "e_w": 0.6, "m_seed": 2.0, "v_kin": 0.65, "eps_kin": 0.61, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-bar...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/mseed2_...
72	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2305.09827.txt", "length": 6000}	[chars 0-6000 of 60884] (more available, next start=6000) Title: A non-linear solution to the S_8 tension II: Analysis of DES Year 3 cosmic shear Authors: Calvin Preston, Alexandra Amon, George Efs...

#	Tool	Arguments	Outcome
73	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.12500.txt", "length": 5000}	[chars 0-5000 of 76057] (more available, next start=5000) Title: Mitigating baryon feedback bias in cosmic shear through a theoretical error covariance in the matter power spectrum Authors: Alessandr...
74	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt", "length": 7000, "start": 42000}	[chars 42000-49000 of 85813] (more available, next start=49000) parameters one by one. 4.1.1 V arying the gas fraction for the thermal AGN model We start by studying the effect of changing the "fgas±...
75	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw2_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=2.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_l...
76	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw4_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=4.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_l...
77	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/mseed2_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=2.0, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_l...

#	Tool	Arguments	Outcome
78	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2305.09827.txt", "start": 48000, "length": 7000}	[chars 48000-55000 of 60884] (more available, next start=55000) Planck prior and include the parameter Amod, the DES and KiDS lensing can be made compatible with Planck, with or without scale cuts. T...
79	local cite	{"arxiv_id": "2410.17109", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/references.bib", "note": "[\"The position of the minimum of the response moves from k = 6...	Cited: The FLAMINGO project: Baryon effects on the matter power spectrum (\cite{arxiv241017109})
80	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt", "length": 9000, "start": 62000}	[chars 62000-71000 of 85813] (more available, next start=71000) these models is relatively broad, as is the baryon fraction in clusters they predict (van Daalen et al. 2020) . Nevertheless, our emulat...
81	board post_finding	{"claim": "Preston, Amon & Efstathiou (2305.09827) find that to reconcile DES Y3 cosmic shear with Planck LCDM, the non-linear suppression needed is weaker than the KiDS-1000 value (A_mod ~ 0.69). It is still more extrem...	Recorded finding f-006.
82	board post_finding	{"claim": "Second mitigation method (Maraio, Hall & Taylor, 2410.12500) : a theoretical-error covariance built into the matter power spectrum, used in place of binary ell scale cuts. For a Euclid-like survey with ell_max ...	Recorded finding f-007.

#	Tool	Arguments	Outcome
83	board post_finding	{"claim": "Approximate comparison only. The HACC emulator gives about 0.8-0.9% suppression at k= 0.5 h/Mpc, about 0.03-0.04% at k=0.1 h/Mpc and at most 2.5% at k=1 h/Mpc (f-001, f-002). This is far below the A_mod-type su...	Recorded finding f-008.
84	board post_question	{"text": "What is the numerical DES Y3 A_mod fitted by Preston et al. (2305.09827)? It is in the body text, chars 6000-48000 of runs/261017_143153/ 2305.09827.txt, which was not read. Also, can 2305.09827 and 2410.12500 b...	Recorded question q-07.
85	board post_finding	{"claim": "Sourced with verbatim quotes in references.bib (key arxiv230902920): Grandis et al. 2309.02920 find a baryonic suppression of the matter power of 0.042 (+0.012/-0.014) at k=1 h/Mpc and 0.184 (+0.026/-0.031) at...	Recorded finding f-009.
86	board post_finding	{"claim": "FLAMINGO (2410.17109, quoted and cited as arxiv241017109) . For the fiducial model, the minimum of P/P_DMO moves from k=6 h/Mpc at z=2 to k=10 h/Mpc at z=0. The departure from unity begins at k~1 h/Mpc at z=2 a...	Recorded finding f-010.
87	board post_finding	{"claim": "Range-of-model comparison of the suppression 1-P_hydro/P_ DMO at z=0. HACC fiducial: 2.2% at k=1 h/Mpc and 8.0% at k=3 h/Mpc. HACC strong AGN: 2.5% and 8.8%. Grandis et al. 2309.02920: 4.2% (+1.2/-1.4) and 18.4...	Recorded finding f-011.

#	Tool	Arguments	Outcome
88	board post_finding	{"claim": "Correction to the f-004 wording. Amon & Efstathiou (2206.11794) show that suppressing the non-linear power with A_mod~0.69 is sufficient to reconcile KiDS-1000 with Planck LCDM. It is one possible route, not t...	Recorded finding f-012.
89	board post_question	{"text": "What is the numerical depth of the FLAMINGO fiducial (and fgas-8sigma) z=0 P/P_DMO minimum near k=10 h/Mpc? It is only shown graphically in Figs. 2 and 5 of 2410.17109, so it needs figure extraction or the publ...	Recorded question q-08.
90	board post_finding	{"claim": "Parameter bounds and meanings for HACC Pk-ratio from describe_parameters: kappa_w is the 'AGN wind coupling / wind efficiency parameter' [2.0, 4.0]; e_w is the 'AGN energy efficiency / wind energy fraction' [0...	Recorded finding f-013.
91	board post_finding	{"claim": "One-at-a-time sensitivity of HACC Pk-ratio at z=0. Except for the parameter varied, all runs use the fiducial point (e_w=0.6, v_kin=0.65, eps_kin=0.61, M_seed=1.3, kappa_w=3, omega_m=0.14, sigma_8=0.8).\nSuppr...	Recorded finding f-014.
92	board post_finding	{"claim": "Restatement of f-002, restricted to the sampled points only. Six in-range points have been sampled: fiducial, strong, weak, kappa_w=2, kappa_w=4 and M_seed=2.\nAt k=1 h/Mpc and z=0, their suppression ranges fr...	Recorded finding f-015.

#	Tool	Arguments	Outcome
93	board post_dead_end	{"what": "Reading the HACC Pk-ratio CSVs beyond k of about 3.73 h/Mpc with read_text_file, in order to get the ratio at k=5 and 10 h/Mpc and the suppression minimum.", "why": "read_text_file has no offset parameter and t...	Recorded dead end x-01.
94	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
95	serve-hydro.sh list_observables	{}	{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...
96	hep-research-aid list_agent_files	{}	[]
97	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
98	hep-research-aid list_datasets	{}	{'count': 0, 'datasets': []}
99	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...
100	hep-research-aid get_dispatch	{}	Dispatch: local execution.

#	Tool	Arguments	Outcome
101	local search_arxiv_ abstracts	{"query": "baryonic suppression matter power spectrum AGN feedback emulator", "max_results": 8}	[0411485v1] Obscured AGN in Multiwavelength Surveys Abstract: Using a simple unification model for AGN, with a fixed ratio of obscured to unobscured AGN of 3: 1, we explain the X-ray, optical and inf...
102	local search_arxiv_ abstracts	{"query": "baryonic feedback weak lensing S8 cosmic shear constraints", "max_results": 6}	[2406.08919v3] Strong gravitational lenses from the Vera C. Rubin Observatory Abstract: Like many areas of astrophysics and cosmology, the Vera C. Rubin Observatory will be transformational for almost...
103	local search_arxiv_ abstracts	{"query": "baryon fraction in haloes predicts power spectrum suppression", "max_ results": 5}	[2509.04552v2] The resummation model in FLAMINGO: precisely predicting matter power suppression from observed halo baryon fractions Abstract: In order to derive unbiased cosmological parameters from S...
104	local search_arxiv_ abstracts	{"max_results": 5, "query": "van Daalen exploring the effects of galaxy formation on matter clustering baryon fraction"}	[1105.1075v3] Quantifying the effect of baryon physics on weak lensing tomography Abstract: We use matter power spectra from cosmological hydrodynamic simulations to quantify the effect of baryon phys...
105	local search_arxiv_ abstracts	{"max_results": 5, "query": "non-linear power spectrum suppression resolves S8 tension Amon Efstathiou"}	[2305.09827v1] A non-linear solution to the S_8 tension II: Analysis of DES Year 3 cosmic shear Abstract: Weak galaxy lensing surveys have consistently reported low values of the S_8 parameter com...

#	Tool	Arguments	Outcome
106	local search_arxiv_ abstracts	{"max_results": 5, "query": "FLAMINGO simulations baryonic effects matter power spectrum stronger feedback"}	[2410.17109v2] The FLAMINGO project: Baryon effects on the matter power spectrum Abstract: The effect of baryon physics associated with galaxy formation onto the large-scale matter distribution of the...
107	local search_arxiv_ abstracts	{"max_results": 5, "query": "CAMELS baryonic effects matter power spectrum feedback parameters"}	[2010.00619v2] The CAMELS project: Cosmology and Astrophysics with MachinE Learning Simulations Abstract: We present the Cosmology and Astrophysics with MachinE Learning Simulations -CAMELS- project...
108	local search_arxiv_ abstracts	{"max_results": 5, "query": "HMcode-2020 baryonic feedback halo model power spectrum"}	[2009.01858v2] HMcode-2020: Improved modelling of non-linear cosmological power spectra with baryonic feedback Abstract: We present an updated version of the HMcode augmented halo model that can be us...
109	local search_arxiv_ abstracts	{"max_results": 5, "query": "kinetic Sunyaev-Zel'dovich weak lensing joint constraint strong baryonic feedback suppression"}	[1902.05141v1] Strong Lensing considerations for the LSST observing strategy Abstract: Strong gravitational lensing enables a wide range of science: probing cosmography; testing dark matter models; un...
110	local search_arxiv_ abstracts	{"max_results": 4, "query": "BACCO emulator baryonification baryonic effects power spectrum"}	[2011.15018v1] The BACCO Simulation Project: A baryonification emulator with Neural Networks Abstract: We present a neural-network emulator for baryonic effects in the non-linear matter power spectrum...

#	Tool	Arguments	Outcome
111	hep-research-aid load_table_data	{"filename": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducial_z0/Pk-ratio_z0.0000_prediction.csv"}	Error: could not convert string 'x' to float64 at row 0, column 1.
112	serve-hydro.sh describe_parameters	{"stat_name": "Pk-ratio"}	{ "status": "success", "message": "7-parameter design space for Pk-ratio: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { "ka...
113	serve-hydro.sh list_observables	{}	{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...
114	serve-hydro.sh list_observables	{}	{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...
115	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153: - 2206.11794.pdf - 2206.11794.txt - 2305.09827.pdf - 2305.09827.txt - 2309.02920.pdf - ...

#	Tool	Arguments	Outcome
116	serve-hydro.sh describe_parameters	{"stat_name": "fGas"}	{ "status": "success", "message": "7-parameter design space for fGas: kappa_w, e_w, M_seed, v_kin, epsilon_kin, omega_m, sigma_8.", "files": [], "metadata": { "parameters": { "kappa_ ...
117	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/strongAGN_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=1.0, m_seed=1.3, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_lab...
118	local download_arxiv_full_text	{"arxiv_id": "2305.09827", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2305.09827.txt
119	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/strongAGN_z0/Pk-ratio_z0.0000_prediction.csv", "start": 17000, "length": 40000}	[chars 17000-39648 of 40115] (more available, next start=39648) 744856247 5.252949, 0.8510089091147397, 0.008317142206033152 5.268652, 0.8507808452225413, 0.008321756313487062 5.284377, 0.8504187991277796...
120	hep-research-aid get_paper_figures	{"paper_id": "2410.17109", "figures": "2,5,6", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127"}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2410.17109 (arxiv-html: https://arxiv.org/html/2410.17109 contains no extractable figures; ar5iv: https://ar5iv.labs.arx...

#	Tool	Arguments	Outcome
121	serve-hydro.sh predict_ observable	{"stat_name": "Pk-ratio", "z": 0, "kappa_w": 2, "e_ w": 1.0, "m_seed": 2.0, "v_kin": 1.2, "eps_kin": 1.2, "omega_m": 0.14, "sigma_8": 0.8, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryoni...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ maxsupp...
122	serve-hydro.sh predict_ observable	{"stat_name": "Pk-ratio", "z": 0, "kappa_w": 2, "e_ w": 1.0, "m_seed": 2.0, "v_kin": 0.1, "eps_kin": 0.02, "omega_m": 0.14, "sigma_8": 0.8, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ramp-combo-baryon...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ maxsupp...
123	serve-hydro.sh predict_ observable	{"stat_name": "Pk-ratio", "z": 0, "kappa_w": 2, "e_ w": 1.0, "m_seed": 0.6, "v_kin": 1.2, "eps_kin": 1.2, "omega_m": 0.14, "sigma_8": 0.8, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryoni...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ maxsupp...
124	serve-hydro.sh predict_ observable	{"stat_name": "Pk-ratio", "z": 0, "kappa_w": 3, "e_ w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "omega_m": 0.14, "sigma_8": 0.8, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ramp-combo-baryo...	{ "status": "success", "message": "Predicted Pk-ratio at z=0.0000 (799 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ fiducia...
125	serve-hydro.sh predict_ observable	{"stat_name": "Pk-ratio", "z": 1.0, "kappa_w": 3, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "omega_m": 0.14, "sigma_8": 0.8, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ramp-combo-bar...	{ "status": "success", "message": "Predicted Pk-ratio at z=1.0000 (799 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ fiducia...

#	Tool	Arguments	Outcome
126	serve-hydro.sh predict_ observable	{"stat_name": "Pk-ratio", "z": 1.0, "kappa_w": 2, "e_w": 1.0, "m_seed": 2.0, "v_kin": 1.2, "eps_kin": 1.2, "omega_m": 0.14, "sigma_8": 0.8, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ramp-combo-baryo...	{ "status": "success", "message": "Predicted Pk-ratio at z=1.0000 (799 bins).", "files": ["/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ maxsupp...
127	local download_arxiv_ full_text	{"arxiv_id": "2512.02954", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/runs/ 261023_143127"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ 2512.02954.txt
128	local download_arxiv_ full_text	{"arxiv_id": "2410.12500", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/runs/ 261023_143127"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ 2410.12500.txt
129	local get_arxiv_ metadata	{"arxiv_id": "2309.02920"}	arXiv ID: 2309.02920 Title: Determining the Baryon Impact on the Matter Power Spectrum with Galaxy Clusters Authors: Sebastian Grandis, Giovanni Arico', Aurel Schneider, Laila Linke Year: 2023 Abstrac...
130	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/runs/ 261017_143153/strongAGN_ z0/Pk-ratio_z0.0000_ prediction.csv", "start": 30500, "length": 10000}	[chars 30500-39921 of 40115] (more available, next start=39921) 0.7971447402970744, 0.00976614659320465 9.540107, 0.7970213220352451, 0.009769374406020503 9.555813, 0.7968189359042359, 0.00977082651626668...

#	Tool	Arguments	Outcome
131	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducial_z0/Pk-ratio_z0.0000_prediction.csv", "start": 0, "length": 20000}	[chars 0-20000 of 40287] (more available, next start=20000) # label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio #...
132	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw2_z0/Pk-ratio_z0.0000_prediction.csv", "start": 0, "length": 300}	[chars 0-300 of 40091] (more available, next start=300) # label: Pk-ratio z=0.0000 (kappa_w=2.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # red...
133	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/weakAGN_z0/Pk-ratio_z0.0000_prediction.csv", "start": 0, "length": 300}	[chars 0-300 of 40016] (more available, next start=300) # label: Pk-ratio z=0.0000 (kappa_w=3.0, e_w=0.2, m_seed=1.3, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # reds...
134	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2305.09827.txt", "length": 6000}	[chars 0-6000 of 60884] (more available, next start=6000) Title: A non-linear solution to the S_8 tension II: Analysis of DES Year 3 cosmic shear Authors: Calvin Preston, Alexandra Amon, George Efs...
135	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2512.02954.txt", "length": 5000}	[chars 0-5000 of 100641] (more available, next start=5000) Title: The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy l...

#	Tool	Arguments	Outcome
136	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.12500.txt", "length": 5000}	[chars 0-5000 of 76057] (more available, next start=5000) Title: Mitigating baryon feedback bias in cosmic shear through a theoretical error covariance in the matter power spectrum Authors: Alessandr...
137	board post_finding	{"claim": "The HACC Pk-ratio emulator has trained redshifts z = 0, 0.1, 0.5, 1.0 and 2.0, with 799 k bins. Its design bounds are kappa_w 2-4, e_w 0.2-1.0, M_seed 0.6-2.0, v_kin 0.1-1.2, eps_kin 0.02-1.2, omega_m 0.12-0.1...	Recorded finding f-016.
138	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z0/Pk-ratio_z0.0000_prediction.csv"}	# label: Pk-ratio z=0.0000 (kappa_w=2.0, e_w=1.0, m_seed=2.0, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 0.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_lab...
139	serve-hydro.sh plot_observable_comparison	{"prediction_files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fiducial_z0/Pk-ratio_z0.0000_prediction.csv", "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo...	{ "status": "success", "message": "Plotted 4 Pk-ratio predictions.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/Pk-ratio_compariso... [Fig. 28]

#	Tool	Arguments	Outcome
140	serve-hydro.sh plot_observable_comparison	{"prediction_files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fiducial_z0/Pk-ratio_z0.0000_prediction.csv", "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo...	{ "status": "success", "message": "Plotted 4 Pk-ratio predictions.", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/Pk-ratio_compariso... [Fig. 28]
141	board post_finding	{"claim": "HACC Pk-ratio (P_hydro/P_grav) at z=0 for the strongAGN point (kappa_w=3.0, e_w=1.0, M_seed=1.3, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) is 0.851 at k=5.25, 0.8205 at k=7.01, 0.7946 at k=10.01 and 0...	Recorded finding f-017.
142	board post_finding	{"claim": "The HACC Pk-ratio emulator k-grid (799 bins) runs from 0.022 to about 12.6 h/Mpc. That answers q-01: the grid reaches 10 h/Mpc but stops at about 12.6 h/Mpc, so the high-k upturn seen in FLAMINGO and OWLS-type...	Recorded finding f-018.
143	board post_finding	{"claim": "At k of about 3.7 h/Mpc and z=0, the fiducial point (kappa_w=3, e_w=0.6, v_kin=0.65, eps_kin=0.61) has a Pk-ratio of 0.8984 (std 0.004), against 0.8909 for strongAGN. At k=1.01 the values are 0.9779 (fiducial)...	Recorded finding f-019.
144	board post_finding	{"claim": "Siegel et al. (arXiv:2512.02954) find from joint eROSITA X-ray gas fractions plus SDSS/DESI+ACT kSZ profiles a matter power suppression of 10 +/- 2% at k = 1 h/Mpc, stronger than the fiducial FLAMINGO and BAHA...	Recorded finding f-020.

#	Tool	Arguments	Outcome
145	board post_finding	{"claim": "Grandis et al. (arXiv:2309.02920) use cluster gas and stellar fractions to infer a suppression of 0.042 (+0.012/-0.014) at k = 1 h/Mpc and 0.184 (+0.026/ -0.031) at k = 3 h/Mpc. They predict, at 97.5% credibili...	Recorded finding f-021.
146	board post_finding	{"claim": "Preston, Amon and Efstathiou (arXiv: 2305.09827) model P_m = P_L + A_mod (P_NL - P_L). Their KiDS-1000 Paper I value is A_mod ≈ 0.69. For DES Y3 the abstract says only that the required suppression is \"not as ...	Recorded finding f-022.
147	board post_finding	{"claim": "Maraio, Hall and Taylor (arXiv:2410.12500) find that for a Euclid-like survey with ell_max = 5000, even multi-parameter baryon models (including HMCode-2020) can bias Omega_ m and S8 significantly. A theoretica...	Recorded finding f-023.
148	board post_dead_end	{"what": "get_paper_ figures on arXiv:2410.17109 (FLAMINGO baryon effects) for Figs. 2, 5, 6 to read the depth of the z=0 minimum.", "why": "Neither the arXiv HTML nor the ar5iv rendering has extractable figures. Reading ...	Recorded dead end x-02.
149	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-baryonic-pk/ runs/261023_143127/ 2305.09827.txt", "start": 6000, "length": 9000}	[chars 6000-15000 of 60884] (more available, next start=15000) . 2023; White et al. 2022; Chen et al. 2022b) should agree with the predictions of the Planck ΛCDM model. As discussed in Paper I there ...

#	Tool	Arguments	Outcome
150	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2305.09827.txt", "length": 6000, "start": 15000}	[chars 15000-21000 of 60884] (more available, next start=21000) ents, resulting in a$\sim 1\sigma$ shift toward a higher value of S8. Variant 3 in Table 2 uses HMCode2020 dark mat- ter model for the non-linear...
151	board post_finding	{"claim": "At z=0 the combined extreme HACC point (kappa_w=2, e_w=1.0, M_seed=2.0, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) gives P_hydro/P_grav = 0.99944+/-0.00003 at k=0.101, 0.9868+/-0.0008 at k=0.510, 0.963...	Recorded finding f-024.
152	board post_dead_end	{"what": "Making two separate comparison plots (the z=0 parameter variants, and z=0 versus z=1) with plot_observable_comparison and the same output_dir.", "why": "Both calls wrote the same file, Pk-ratio_comparison.png, ...	Recorded dead end x-03.
153	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z1/Pk-ratio_z1.0000_prediction.csv"}	# label: Pk-ratio z=1.0000 (kappa_w=2.0, e_w=1.0, m_seed=2.0, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # redshift: 1.0 # x_label: $k \, [h\,\mathrm{Mpc}^{-1}]$ # y_lab...
154	serve-hydro.sh predict_observable	{"stat_name": "fGas", "kappa_w": 3.0, "e_w": 0.6, "m_seed": 1.3, "v_kin": 0.65, "eps_kin": 0.61, "omega_m": 0.14, "sigma_8": 0.8, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryoni...	{ "status": "success", "message": "Predicted fGas at z=0.0000 (3 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_fiducial...

#	Tool	Arguments	Outcome
155	serve-hydro.sh predict_observable	{"stat_name": "fGas", "kappa_w": 3.0, "e_w": 1.0, "m_seed": 1.3, "v_kin": 1.2, "eps_kin": 1.2, "omega_m": 0.14, "sigma_8": 0.8, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-...	{ "status": "success", "message": "Predicted fGas at z=0.0000 (3 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_strong/f...
156	serve-hydro.sh predict_observable	{"stat_name": "fGas", "kappa_w": 3.0, "e_w": 0.2, "m_seed": 1.3, "v_kin": 0.1, "eps_kin": 0.02, "omega_m": 0.14, "sigma_8": 0.8, "z": 0, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic...	{ "status": "success", "message": "Predicted fGas at z=0.0000 (3 bins).", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_weak/fGa...
157	board post_finding	{"claim": "Answers q-07. Preston, Amon and Efstathiou (arXiv:2305.09827, Eq. 3) find that Planck LCDM is compatible with DES Y3 cosmic shear if non-linear power is suppressed with A_mod = 0.858 +/- 0.052 (no scale cuts, ...	Recorded finding f-025.
158	board post_finding	{"claim": "fGas (cluster gas fraction vs M500c, 3 mass bins) was predicted at z=0 for the fiducial, strongAGN and weakAGN points. The CSVs are written, but their values were not read because the tool budget ran out, so t...	Recorded finding f-026.
159	board post_dead_end	{"what": "Loading the HACC Pk-ratio prediction CSVs with load_table_data", "why": "The CSV header row 'x,y_mean,y_std' after the '#' comments is not skipped, and the tool fails with \ "could not convert string 'x' to floa...	Recorded dead end x-04.

#	Tool	Arguments	Outcome
160	board post_question	{"text": "What are the fGas values at M500c of about 1e14 Msun/h in the three fGas CSVs under runs/261023_143127/fgas_*, and do they fall monotonically as the k=1 and k=3 suppression grows (fiducial 0.978/0.920, strongAG...	Recorded question q-12.
161	local cite	{"arxiv_id": "2305.09827", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/references.bib", "note": "[\"The non-linear suppression of the matter power spectrum requ...	Cited: A non-linear solution to the S_8 tension II: Analysis of DES Year 3 cosmic shear (\cite{arxiv230509827})
162	local cite	{"arxiv_id": "2410.12500", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/references.bib", "note": "[\"our theoretical error reduces the bias inΩm and\"]"}	Cited: Mitigating baryon feedback bias in cosmic shear through a theoretical error covariance in the matter power spectrum (\cite{arxiv241012500})
163	local cite	{"arxiv_id": "2512.02954", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/references.bib", "note": "[\"The inferred gas profiles are more extended and the power su...	Cited: The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy lensing (\cite{arxiv251202954})

#	Tool	Arguments	Outcome
164	local cite	{"arxiv_id": "2309.02920", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/references.bib", "note": "[\"This puts into question if baryon feedback is the driving fa...	Cited: Determining the Baryon Impact on the Matter Power Spectrum with Galaxy Clusters (\cite{arxiv230902920})
165	board post_finding	{"claim": "For the combined extreme HACC point, suppression weakens strongly from z=0 to z=1. At k=1.013 h/Mpc the ratio rises from 0.9638+/-0.0020 at z=0 to 0.9781+/-0.0018 at z=1, so the suppression drops from 3.6% to ...	Recorded finding f-027.
166	board post_finding	{"claim": "Four validated cite entries (arxiv230509827, arxiv241012500, arxiv251202954, arxiv230902920) were written to /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/reference...	Recorded finding f-028.
167	board post_question	{"text": "q-08 is still open. Figure extraction for FLAMINGO arXiv:2410.17109 failed (no HTML figures), so the depth of the z=0 minimum needs reading from the full text, via download_arxiv_full_text and read_arxiv_chunk ...	Recorded question q-13.
168	local download_arxiv_full_text	{"arxiv_id": "2410.12500", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.12500.txt

#	Tool	Arguments	Outcome
169	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_fiducial/fGas_z0.0000_prediction.csv", "length": 2000}	[chars 0-477 of 481] (more available, next start=477) # label: fGas z=0.0000 (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.14, sigma_8=0.8) # observable: fGas # redshift: 0.0...
170	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_strong/fGas_z0.0000_prediction.csv", "length": 2000}	[chars 0-476 of 480] (more available, next start=476) # label: fGas z=0.0000 (kappa_w=3.0, e_w=1.0, m_seed=1.3, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) # observable: fGas # redshift: 0.0 #...
171	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_weak/fGas_z0.0000_prediction.csv", "length": 2000}	[chars 0-475 of 479] (more available, next start=475) # label: fGas z=0.0000 (kappa_w=3.0, e_w=0.2, m_seed=1.3, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) # observable: fGas # redshift: 0.0 ...
172	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z0"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z0: - Pk-ratio_z0.0000_prediction.csv
173	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowkin_z0"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowkin_z0: - Pk-ratio_z0.0000_prediction.csv

#	Tool	Arguments	Outcome
174	local download_arxiv_full_text	{"arxiv_id": "2410.17109", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.17109.txt
175	local download_arxiv_full_text	{"arxiv_id": "2512.02954", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2512.02954.txt
176	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z0/Pk-ratio_z0.0000_prediction.csv", "start": 13000, "length": 12000}	[chars 13000-25000 of 40083] (more available, next start=25000) 285429 3.996619, 0.8283594039852574, 0.00794776027895483 4.012363, 0.8277671159583021, 0.00796877927206965 4.028096, 0.8271063929032664, 0.00...
177	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowkin_z0/Pk-ratio_z0.0000_prediction.csv", "start": 0, "length": 700}	[chars 0-700 of 40101] (more available, next start=700) # label: Pk-ratio z=0.0000 (kappa_w=2.0, e_w=1.0, m_seed=2.0, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # reds...
178	local download_arxiv_full_text	{"arxiv_id": "2309.02920", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2309.02920.txt

#	Tool	Arguments	Outcome
179	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.12500.txt", "start": 40000, "length": 15000}	[chars 40000-55000 of 76057] (more available, next start=55000) ly dependent on the combination of spectra considered, with low redshift bins having a smaller error than those at high redshift (which...
180	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowkin_z0/Pk-ratio_z0.0000_prediction.csv", "start": 700, "length": 11800}	[chars 700-12500 of 40101] (more available, next start=12500) , 1.0002173508631655, 7.272374987438343e-05 0.1644056, 1.0001507404372716, 8.975520432802089e-05 0.1799158, 1.0002057129329116, 0.0001064335291...
181	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowM_z0/Pk-ratio_z0.0000_prediction.csv", "start": 0, "length": 12500}	[chars 0-12500 of 40004] (more available, next start=12500) # label: Pk-ratio z=0.0000 (kappa_w=2.0, e_w=1.0, m_seed=0.6, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) # observable: Pk-ratio # r...
182	board post_finding	{"claim": "Corrected deficit significance for the HACC extreme point (f-024), with emulator and published errors combined in quadrature and the published error taken on the side facing the HACC value. Against Grandis et ...	Recorded finding f-029.
183	board post_finding	{"claim": "Full texts of FLAMINGO (2410.17109), Siegel et al. (2512.02954) and Grandis et al. (2309.02920) were downloaded to /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/241...	Recorded finding f-030.

#	Tool	Arguments	Outcome
184	board post_question	{"text": "Using read_arxiv_chunk on the downloaded .txt files in runs/261023_143127/, can someone confirm verbatim the 10+/-2% and 5+/-4% values (2512.02954) and the 0.042 and 0.184 values (2309.02920), and quote the FLA...	Recorded question q-14.
185	board post_finding	{"claim": "Maraio, Hall and Taylor (arXiv:2410.12500, Table 3) work with an idealised Euclid-like (Stage-IV) cosmic-shear survey at ell_max = 5000 and use Illustris as the ground truth. When the one-parameter HMCode bary...	Recorded finding f-031.
186	board post_finding	{"claim": "For the combined extreme point, the HACC maxsupp z=0 Pk-ratio keeps falling past k=3.7 h/Mpc: 0.8284+/-0.0079 at k=4.00, 0.7942+/-0.0089 at k=5.00, 0.7680+/-0.0095 at k=6.0, 0.7479+/-0.0100 at k=7.01 and 0.735...	Recorded finding f-032.
187	board post_question	{"text": "Two values are still unread. First, the maxsupp_z0 S(k) at k=0.2-0.3 h/Mpc (CSV chars about 0-3000). From f-024 it lies between 0.9994 at k=0.1 and 0.987 at k=0.51, so it is likely below 1%, which is far weaker...	Recorded question q-15.
188	board post_finding	{"claim": "The HACC fGas emulator at z=0 has 3 M500c bins: 5.01e13, 7.94e13 and 1.26e14 Msun/h. In the bin nearest 1e14 (1.26e14 Msun/h), f_gas = 0.1214+/-0.0010 for fiducial, 0.1221+/-0.0019 for strongAGN and 0.1414+/-0...	Recorded finding f-033.

#	Tool	Arguments	Outcome
189	board post_finding	{"claim": "Partial answer to q-12. Only two fGas points have S(k) read so far. Fiducial: f_gas(1.26e14) = 0.1214+/-0.0010, with S(k=1) = 0.9779 and S(k=3) = 0.9197. StrongAGN: f_gas = 0.1221+/-0.0019, with S(k=1) = 0.974...	Recorded finding f-034.
190	board post_finding	{"claim": "The HACC maxsupp_lowkin point at z=0 (kappa_w=2, e_w=1.0, M_seed=2.0, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) shows essentially no suppression. P_hydro/P_grav is 1.0002 at k=0.212, 1.0004+/-0.0007 ...	Recorded finding f-035.
191	board post_finding	{"claim": "The HACC maxsupp_lowM point at z=0 (kappa_w=2, e_w=1.0, M_seed=0.6, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) suppresses more than the combined maxsupp point (M_seed=2.0). P_hydro/P_grav is 0.9967 at ...	Recorded finding f-036.
192	board post_question	{"text": "Still open from d-11: (a) weakAGN S(k=1) and S(k=3) (runs/261017_143153/weakAGN_z0) , needed for a three-point f_gas-suppression test; (b) fiducial_z1 S(k=1) and S(k=3) (runs/261023_143127/fiducial_z1); (c) a ve...	Recorded question q-16.

A.5 Files produced

- 2309.07959.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.07959.txt
- 2410.17109.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.17109.txt
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/fiducial_z0/Pk-ratio_z0.0000_prediction.csv
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/

hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/strongAGN_z0/Pk-ratio_z0.0000_prediction.csv

- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/weakAGN_z0/Pk-ratio_z0.0000_prediction.csv
- 2206.11794.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2206.11794.txt
- 2512.02954.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2512.02954.txt
- 2309.02920.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2309.02920.txt
- 2305.09827.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2305.09827.txt
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw2_z0/Pk-ratio_z0.0000_prediction.csv
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/kappaw4_z0/Pk-ratio_z0.0000_prediction.csv
- 2410.12500.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/2410.12500.txt
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/mseed2_z0/Pk-ratio_z0.0000_prediction.csv
- 2305.09827.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2305.09827.txt
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z0/Pk-ratio_z0.0000_prediction.csv
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowkin_z0/Pk-ratio_z0.0000_prediction.csv
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_lowM_z0/Pk-ratio_z0.0000_prediction.csv
- Pk-ratio_z0.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fiducial_z0/Pk-ratio_z0.0000_prediction.csv
- Pk-ratio_z1.0000_prediction.csv (csv, predict_observable): /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/fiducial_z1/Pk-ratio_z1.0000_prediction.csv

- `Pk-ratio_z1.0000_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/maxsupp_z1/Pk-ratio_z1.0000_prediction.csv`
- `2512.02954.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2512.02954.txt`
- `2410.12500.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.12500.txt`
- `fGas_z0.0000_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testi projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_fiducial/fGas_z0.0000_prediction.csv`
- `fGas_z0.0000_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testi projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_strong/fGas_z0.0000_prediction.csv`
- `fGas_z0.0000_prediction.csv` (csv, predict_observable): `/Users/nramachandra/hep-genesis-testi projects/ramp-combo-baryonic-pk/runs/261023_143127/fgas_weak/fGas_z0.0000_prediction.cs`
- `2410.17109.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.17109.txt`
- `2309.02920.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2309.02920.txt`

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
`d-01`	closed	Use the HACC hydro emulator to predict the matter power spectrum ratio P_hydro/P_DMO at z=0 for k from about 0.1 to 10 h/Mpc at two or three parameter points: fiducial, strong AGN feedback and weak AGN feedback. Report the depth and wavenumber of the suppression minimum and the suppression at k=1 h/Mpc.	I ran three HACC Pk-ratio (P_hydro/P_grav) predictions at z=0 with cosmology held fixed (omega_m=0.14, sigma_8=0.8, kappa_w=3, M_seed=1.3). Two success criteria are not fully met: values at k=5-10 h/Mpc and the location and depth of the suppression minimum were not extracted. The text reader cuts each 799-bin CSV off near k=3.7 h/Mpc, and load_table_data rejects the CSV header row (posted as q-06)

Id	Status	Goal	Outcome
d-02	closed	Build a cited literature synthesis covering two topics: (a) the AGN feedback physics that causes baryonic suppression, and (b) its impact on weak-lensing cosmology, including the S8 bias and mitigation methods. Extract numerical benchmarks to compare against the emulator results.	This direction is only partly done. The literature synthesis is posted as finding f-003 and rests on the abstracts and introductions of five full texts. These are saved under /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/ as 2410.17109.txt, 2512.02954.txt, 2309.07959.txt, 2206.11794.txt and 2309.02920.txt. Mechanism, from those texts: - Energy injected
d-03	closed	Robustness and physics cross-checks with the emulator. Test sensitivity of the z=0 Pk-ratio to each feedback parameter separately and to cosmology. Check redshift evolution of the suppression minimum. Link suppression to the group and cluster gas fraction using the emulator's fGas observable.	
d-04	closed	Cited literature synthesis of the impact of baryonic suppression on weak-lensing cosmology: the S8 bias if baryons are ignored, the suppression amplitude needed to reconcile shear with Planck, and mitigation methods.	Three papers were read and cited with verbatim quotes in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/references.bib. (1) Amon & Efstathiou (2206.11794): KiDS-1000 cosmic shear is consistent with Planck LCDM if non-linear power is suppressed with A_mod ~ 0.69. The suppression must extend to k ~ 0.2 h/Mpc, which is well outside the BAHAMAS range. (2)

Id	Status	Goal	Outcome
d-05	closed	Find out which HACC emulator parameters control the z=0 Pk-ratio suppression, and resolve the curve beyond k=3.7 h/Mpc. This is a budget-limited replacement for d-03.	This direction is partly done. The parameter bounds, the sensitivity ranking and the 5%/10% check were completed. The suppression minimum is not resolved above k=3.73 h/Mpc, and the comparison plot was not made. Results are in f-013, f-014 and f-015, and the dead end is x-01. Parameter bounds, quoted from describe_parameters (f-013): - kappa_w is the 'AGN wind coupling / wind efficiency parameter
d-06	closed	Close the literature sourcing gaps and cross-check the HACC suppression amplitude against published calibrated values at k=1 and k=3 h/Mpc.	Two sourcing gaps are closed, with verbatim quotes stored in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/references.bib; the depth of the FLAMINGO z=0 minimum is still open. (1) 2309.02920 (key arxiv230902920), finding f-009: suppression of 0.042 (+0.012/-0.014) at k=1 h/Mpc and 0.184 (+0.026/-0.031) at k=3 h/Mpc in the fiducial model, or 0.049 and 0.
d-07	blocked	Source the cosmic-shear impact of baryonic suppression: the affected ell or theta range, the DES Y3 A_mod, and a second mitigation method.	Partly done. Three success criteria are not met: (1) the DES Y3 A_mod number was not found; (2) nothing was run through cite, so 2305.09827 and 2410.12500 are not in references.bib; (3) no theta range was sourced. What was established, from full texts saved in /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261017_143153/ (2305.09827.txt, 2410.12500.txt): (a) Finding f

Id	Status	Goal	Outcome
d-08	closed	Find the strongest suppression the HACC emulator can produce at z=0. Then measure how the Pk-ratio evolves from z=0 to z=1 for the fiducial point and the maximum-suppression point.	The direction is mostly done, with three gaps. First, the fiducial z=0 and z=1 values were written to CSV but not read, so the z=0 vs z=1 comparison so far covers only one parameter point. Second, the maxsupp_lowkin and maxsupp_lowM variants were plotted but their values were not read, so it is still untested whether the kinetic parameters or M_seed act non-monotonically. Third, only one of the tw
d-09	closed	Resolve the HACC Pk-ratio above k=3.7 h/Mpc (q-06, x-01), giving values at k=5 and 10 and the location and depth of the minimum. Also link the suppression to the group-scale gas fraction using the emulator's fGas observable.	The high-k part of this direction is resolved. The fGas link is only half done: the predictions were run but never read. 1. Reading past k=3.7 h/Mpc. load_table_data cannot read the Pk-ratio CSVs because it does not skip the header row 'x,y_mean,y_std' and has no skiprows option (dead end x-04, tc-...-19). The files can be read in full with read_arxiv_chunk using start/length offsets. 2. Emulato
d-10	closed	Close the remaining literature gaps: the DES Y3 A_mod value (q-07), the depth of the FLAMINGO z=0 minimum (q-08), and cite entries for 2305.09827, 2410.12500, 2512.02954 and 2309.02920.	Two of the three targets are met. The FLAMINGO z=0 minimum depth (q-08) is still open. q-07 answered (f-025): Preston, Amon and Efstathiou (arXiv:2305.09827, Eq. 3) give A_mod = 0.858 +/- 0.052 for DES Y3 with no scale cuts and a Planck prior on S8 and Omega_m. The verbatim quote is 'Amod = 0.858± 0.052, DES Y3 (no scale cuts)' (tc-...-57). For comparison, KiDS-1000 in Paper I gave A_mod ≈ 0.69 (

Id	Status	Goal	Outcome
d-11	closed	Read the HACC outputs already written but never read (fGas for fiducial, strongAGN and weakAGN; maxsupp_lowkin_z0; maxsupp_lowM_z0; fiducial_z1). Pair f_gas near M500c ~ 1e14 Msun with S(k=1) and S(k=3) to test the van Daalen-type relation, and establish whether the design range can reach the eROSITA+kSZ level.	Direction partly met. The f_gas values and the two maxsupp variants were read; the weakAGN pairing and the fiducial z=0 vs z=1 comparison were not. (1) fGas at z=0 (f-033), in the bin nearest 1e14 Msun/h (M500c = 1.26e14 Msun/h): fiducial 0.1214+/-0.0010, strongAGN 0.1221+/-0.0019, weakAGN 0.1414+/-0.0023. (2) Gas-fraction vs suppression (f-034): this pairing is not met. Only 2 points have S(k) va
d-12	blocked	Verify the contested literature numbers verbatim and close the FLAMINGO gap. Quote 10+/-2% and 5+/-4% from 2512.02954, quote 0.042 and 0.184 at k=1 and 3 from 2309.02920, and quote the FLAMINGO z=0 S(k) at k=1 and the depth and location of its minimum from 2410.17109.	Only the error-budget part of this direction is done. No text was read, so f-020 and f-021 are still not confirmed verbatim and no FLAMINGO S(k) value has been quoted. (1) Full texts of 2410.17109 (FLAMINGO), 2512.02954 and 2309.02920 were downloaded to /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/ as 2410.17109.txt, 2512.02954.txt and 2309.02920.txt (
d-13	blocked	Make the weak-lensing impact quantitative. Extract one numerical S8 (or Omega_m) bias from ignoring baryons, with its survey configuration. Compare HACC maxsupp S(k) at k=0.2-0.5 with the Preston et al. requirement that suppression extend to k~0.2 h/Mpc and with the size of A_mod.	Only part of this direction got done. I have one numerical cosmology bias from ignoring baryons, but it is an Omega_m bias, not an S8 bias, and I did not read the HACC values at k=0.2-0.3 h/Mpc. What was established: 1. Omega_m bias from Maraio, Hall and Taylor (arXiv:2410.12500, Table 3), read verbatim (f-031; tc-...-86). Survey configuration: an idealised Euclid-like Stage-IV cosmic-shear surv

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	medium	HACC emulator Pk-ratio (P_hydro/P_grav) at z=0, cosmology fixed (omega_m=0.14, sigma_8=0.8, kappa_w=3, M_seed=1.3). Fiducial (e_w=0.6, v_kin=0.65, eps_kin=0.61): 0.99968 at k=0.101, 0.99197 at k=0.510, 0.97832+/-0.00096 at k=0.997 (2.2% suppression), 0.95104+/-0.00229 at k=2.00, 0.91968+/-0.00346 at k=3.01, 0.8979 at k=3.71 h/Mpc. Strong AGN (e_w=1.0, v_kin=1.2, eps_kin=1.2): 0.99962, 0.99062, 0.97506+/-0.00175 (2.5%), 0.94463+/-0.00405, 0.91207+/-0.00596, 0.8904 at the same k. Suppression still deepens at k=3.7 h/Mpc in both runs, so the minimum lies above k=3.7. At k=1 h/Mpc both runs are wi	tc-261017_1-w1-d-01-26, tc-261017_1-w1-d-01-27, tc-261017_1-w1-d-01-31, tc-261017_1-w1-d-01-37, tc-261017_1-w1-d-01-34, ... Fig. 8, 9
f-002	medium, **contested**	The HACC Pk-ratio at z=0 for the weak-AGN point (kappa_w=3, e_w=0.2, M_seed=1.3, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) shows no suppression. Instead it shows an enhancement: 1.00009 at k=0.101, 1.00096 at k=0.510 and 1.0023+/-0.0025 at k=0.997 h/Mpc. The ratio peaks at about 1.0097+/-0.0072 near k of about 2.4 h/Mpc and falls back to 1.0012+/-0.0105 at k=3.71 h/Mpc. Across the full design range of the AGN and kinetic parameters, the k=1 h/Mpc suppression spans only about -0.2% to +2.5%. The emulator is therefore consistent with HSC-XXL (5+/-4%) at the strong end but cannot reach	tc-261017_1-w1-d-01-28, tc-261017_1-w1-d-01-49, tc-261017_1-w1-d-01-31, tc-261017_1-w1-d-01-37 Fig. 8
f-003	high, **contested**	Literature synthesis from full texts that were read: (1) Mechanism: energy from active galactic nuclei (AGN) moves gas relative to the dark matter. This suppresses the nonlinear matter power by tens of percent compared with dark-matter-only runs (2512.02954). Below about 10 Mpc, AGN redistribute baryons outward, while on smaller scales star formation concentrates baryons in halo centres (2309.02920). In FLAMINGO, lower group and cluster gas fractions, collimated jets instead of thermal winds, and lower stellar mass at fixed gas fraction all strengthen the response. As z falls, the response gro	tc-261017_1-w1-d-02-43, tc-261017_1-w1-d-02-44, tc-261017_1-w1-d-02-45, tc-261017_1-w1-d-02-46, tc-261017_1-w1-d-02-47

Id	Confidence	Claim	Evidence (tool calls, figures)
f-004	high	Literature (quoted, cited): Amon & Efstathiou (2206.11794) find KiDS-1000 xi_pm is consistent with Planck LCDM only if non-linear power is suppressed with A_mod ~ 0.69, extending to k ~ 0.2 h/Mpc, outside the BAHAMAS range; FLAMINGO (2309.07959) finds baryonic effects across its feedback variants are not large enough to remove the S8 tension; Siegel et al. (2512.02954) find eROSITA+kSZ prefer 10+/-2% suppression at k=1 h/Mpc (HSC-XXL 5+/-4%), only BAHAMAS 8.0 and FLAMINGO fgas-8sigma agree, simulations differ by >10% at k=1 h/Mpc, and in an LSST-Y1 mock, ignoring baryons significantly biases S	tc-261017_1-w1-d-04-36, tc-261017_1-w1-d-04-41, tc-261017_1-w1-d-04-40, tc-261017_1-w1-d-04-42, tc-261017_1-w1-d-04-50, ...
f-005	medium	This is a qualitative comparison only, and the d-01 HACC emulator values were not on the board when it was posted. Any emulator suppression at k=1 h/Mpc should be read against these benchmarks: about 5% (HSC-XXL, consistent with fiducial FLAMINGO and BAHAMAS), 10+/-2% (eROSITA+kSZ, strong feedback), and the much stronger, larger-scale suppression (A_mod ~ 0.69, reaching k~0.2 h/Mpc) that Amon & Efstathiou need to reconcile KiDS with Planck. A suppression of 10% or less at k=1 h/Mpc would, per FLAMINGO 2309.07959, not by itself remove the S8 tension.	tc-261017_1-w1-d-04-40, tc-261017_1-w1-d-04-41, tc-261017_1-w1-d-04-36
f-006	high	Preston, Amon & Efstathiou (2305.09827) find that to reconcile DES Y3 cosmic shear with Planck LCDM, the non-linear suppression needed is weaker than the KiDS-1000 value (A_mod ~ 0.69). It is still more extreme than recent hydrodynamic simulations predict, and it must extend into the mildly non-linear regime, to k ~ 0.2 h/Mpc. The DES Y3 A_mod value itself was not in the text I read (abstract and conclusions), so it remains unsourced.	tc-261017_1-w2-d-07-72, tc-261017_1-w2-d-07-78

Id	Confidence	Claim	Evidence (tool calls, figures)
f-007	high, **contested**	Second mitigation method (Maraio, Hall & Taylor, 2410.12500): a theoretical-error covariance built into the matter power spectrum, used in place of binary ell scale cuts. For a Euclid-like survey with ell_max = 5000, even multi-parameter baryon models can introduce significant bias. The theoretical error reduces the bias in Omega_m and S8 to acceptable levels with only a modest increase in variance, and it removes the need to choose an ell_max for each bin. This places the baryon-sensitive shear range up to ell ~ 5000.	tc-261017_1-w2-d-07-73
f-008	medium, **contested**	Approximate comparison only. The HACC emulator gives about 0.8-0.9% suppression at k=0.5 h/Mpc, about 0.03-0.04% at k=0.1 h/Mpc and at most 2.5% at k=1 h/Mpc (f-001, f-002). This is far below the A_mod-type suppression that Amon-Efstathiou and Preston et al. require. That suppression scales the whole non-linear part of P(k) (A_mod ~ 0.69 for KiDS, weaker but still beyond simulations for DES) and reaches down to k ~ 0.2 h/Mpc. Within its design range, the emulator cannot approach the suppression needed to reconcile shear with Planck. A projected C_ell or xi_+ forecast is future work, since no t	tc-261017_1-w2-d-07-72, tc-261017_1-w1-d-01-26, tc-261017_1-w1-d-01-27
f-009	high	Sourced with verbatim quotes in references.bib (key arxiv230902920): Grandis et al. 2309.02920 find a baryonic suppression of the matter power of 0.042 (+0.012/-0.014) at k=1 h/Mpc and 0.184 (+0.026/-0.031) at k=3 h/Mpc in their fiducial baryon-correction model. Their second model gives 0.049 (+0.016/-0.012) and 0.179 (+0.018/-0.020). At 97.5 percent credibility the impact is below 1% for k<0.37 h/Mpc. They exclude some hydrodynamical simulations as having feedback that is too strong or too weak.	tc-261017_1-w2-d-06-63, tc-261017_1-w2-d-06-67

Id	Confidence	Claim	Evidence (tool calls, figures)
f-010	high	FLAMINGO (2410.17109, quoted and cited as arxiv241017109). For the fiducial model, the minimum of P/P_DMO moves from k=6 h/Mpc at z=2 to k=10 h/Mpc at z=0. The departure from unity begins at k~1 h/Mpc at z=2 and at k~0.1 h/Mpc at z=0. At z=0, across thermal-AGN fgas variants from -10 sigma to +4 sigma, the minimum stays at k=8-10 h/Mpc. The fgas parameter mainly changes the normalisation of the response, and the response saturates below fgas-8 sigma (an emulator extrapolation). The jet model has a stronger response at negative fgas but a similar k-range and minimum position. The text gives no	tc-261017_1-w2-d-06-71, tc-261017_1-w2-d-06-74, tc-261017_1-w2-d-06-80, tc-261017_1-w2-d-06-79
f-011	medium, **contested**	Range-of-model comparison of the suppression 1-P_hydro/P_DMO at z=0. HACC fiducial: 2.2% at k=1 h/Mpc and 8.0% at k=3 h/Mpc. HACC strong AGN: 2.5% and 8.8%. Grandis et al. 2309.02920: 4.2% (+1.2/-1.4) and 18.4% (+2.6/-3.1), or 4.9% and 17.9% in their alternative model. HSC-XXL: 5+/-4% at k=1 only. eROSITA+kSZ: 10+/-2% at k=1 only (2512.02954). At k=1, HACC is about 1.2-1.4 sigma below 2309.02920 and consistent with HSC-XXL. At k=3, HACC is low by a factor of about 2.1-2.3, which is about 3.1-3.4 sigma below the 2309.02920 lower error bar, even at the edge of the HACC design range. This gap is	tc-261017_1-w1-d-01-31, tc-261017_1-w1-d-01-37, tc-261017_1-w2-d-06-63, tc-261017_1-w1-d-04-41
f-012	medium	Correction to the f-004 wording. Amon & Efstathiou (2206.11794) show that suppressing the non-linear power with A_mod~0.69 is sufficient to reconcile KiDS-1000 with Planck LCDM. It is one possible route, not the only one. On mitigation, Siegel et al. (2512.02954) find in their LSST-Y1 mock that an uninformed baryonification model removes the S8 bias but roughly doubles the S8 uncertainty (precision halved). A kSZ+eROSITA-calibrated prior recovers about half of that lost precision.	tc-261017_1-w1-d-04-36, tc-261017_1-w1-d-04-50, tc-261017_1-w1-d-04-51, tc-261017_1-w1-d-04-52

Id	Confidence	Claim	Evidence (tool calls, figures)
f-013	high	Parameter bounds and meanings for HACC Pk-ratio from describe_parameters: kappa_w is the 'AGN wind coupling / wind efficiency parameter' [2.0, 4.0]; e_w is the 'AGN energy efficiency / wind energy fraction' [0.2, 1.0]; M_seed is the black hole seed mass in units of 1e6 Msun [0.6, 2.0]; v_kin is the kinetic feedback velocity in units of 1e4 km/s [0.1, 1.2]; epsilon_kin is the kinetic feedback efficiency in units of 10 [0.02, 1.2]; omega_m is the physical matter density, omega_m = Omega_m h^2 [0.12, 0.155]; sigma_8 spans [0.7, 0.9]. The design has 110 simulations in 400 Mpc/h boxes.	tc-261017_1-w2-d-05-62
f-014	high	One-at-a-time sensitivity of HACC Pk-ratio at z=0. Except for the parameter varied, all runs use the fiducial point (e_w=0.6, v_kin=0.65, eps_kin=0.61, M_seed=1.3, kappa_w=3, omega_m=0.14, sigma_8=0.8). Suppression, 1-P_hydro/P_grav (emulator 1-sigma in brackets): - kappa_w=2: 2.92% (0.15%) at k=0.997; 10.93% (0.54%) at k=3.007; 13.66% (0.64%) at k=3.714 h/Mpc. - kappa_w=4: 1.31% (0.13%) at k=0.997; 4.95% (0.46%) at k=3.007; 6.60% (0.54%) at k=3.714 h/Mpc. - M_seed=2.0: 2.42% (0.14%) at k=0.997; 9.03% (0.49%) at k=3.007; 11.40% (0.57%) at k=3.714 h/Mpc. - Fiducial, from f-001: 2.17% (0.10%) at	tc-261017_1-w2-d-05-68, tc-261017_1-w2-d-05-69, tc-261017_1-w2-d-05-70, tc-261017_1-w2-d-05-75, tc-261017_1-w2-d-05-76, ...
f-015	medium	Restatement of f-002, restricted to the sampled points only. Six in-range points have been sampled: fiducial, strong, weak, kappa_w=2, kappa_w=4 and M_seed=2. At k=1 h/Mpc and z=0, their suppression ranges from -0.2% to 2.9%; the maximum is 2.92+/-0.15% at kappa_w=2. None reaches 5% or 10%. The most suppressed point: - is consistent with HSC-XXL (5+/-4%) within about 0.5 sigma; - falls short of eROSITA+kSZ (10+/-2%, 2512.02954) by about 3.5 sigma. At k=3 h/Mpc the largest sampled suppression is 10.9+/-0.5% (kappa_w=2). This is below the 18.4 (+2.6/-3.1)% from 2309.02920 by about 2.4 sigma, usi	tc-261017_1-w2-d-05-75, tc-261017_1-w2-d-05-76, tc-261017_1-w2-d-05-77, tc-261017_1-w1-d-01-31, tc-261017_1-w1-d-01-37

Id	Confidence	Claim	Evidence (tool calls, figures)
f-016	high	The HACC Pk-ratio emulator has trained redshifts z = 0, 0.1, 0.5, 1.0 and 2.0, with 799 k bins. Its design bounds are kappa_w 2-4, e_w 0.2-1.0, M_seed 0.6-2.0, v_kin 0.1-1.2, eps_kin 0.02-1.2, omega_m 0.12-0.155 and sigma_8 0.7-0.9. Predictions were written for maxsupp, maxsupp_lowkin and maxsupp_lowM at z=0, and for fiducial and maxsupp at z=0 and at z=1.	tc-261023_1-w1-d-08-18, tc-261023_1-w1-d-08-21, tc-261023_1-w1-d-08-29, tc-261023_1-w1-d-08-30, tc-261023_1-w1-d-08-31, ...
f-017	high	HACC Pk-ratio (P_hydro/P_grav) at z=0 for the strongAGN point (kappa_w=3.0, e_w=1.0, M_seed=1.3, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) is 0.851 at k=5.25, 0.8205 at k=7.01, 0.7946 at k=10.01 and 0.7893 at k=12.55 h/Mpc (the last tabulated k). The curve flattens to a plateau of about 0.789-0.790 for k of about 11.6-12.6 h/Mpc, so there is no resolved upturn: the minimum is at or just beyond k_max of about 12.6 h/Mpc and has a depth of about 21%.	tc-261023_1-w1-d-09-27, tc-261023_1-w1-d-09-37
f-018	high	The HACC Pk-ratio emulator k-grid (799 bins) runs from 0.022 to about 12.6 h/Mpc. That answers q-01: the grid reaches 10 h/Mpc but stops at about 12.6 h/Mpc, so the high-k upturn seen in FLAMINGO and OWLS-type runs is not covered. For the strongAGN point the ratio is still decreasing slowly (by less than 0.001 per 0.1 h/Mpc) at the edge.	tc-261023_1-w1-d-09-20, tc-261023_1-w1-d-09-37
f-019	high, **contested**	At k of about 3.7 h/Mpc and z=0, the fiducial point (kappa_w=3, e_w=0.6, v_kin=0.65, eps_kin=0.61) has a Pk-ratio of 0.8984 (std 0.004), against 0.8909 for strongAGN. At k=1.01 the values are 0.9779 (fiducial) and 0.9746 (strongAGN), and at k=3.01 they are 0.9197 and 0.9121. Strong kinetic feedback therefore deepens the suppression by only about 0.3-0.8 percentage points over k=1-3.7 h/Mpc.	tc-261023_1-w1-d-09-38, tc-261023_1-w1-d-09-25
f-020	high, **contested**	Siegel et al. (arXiv:2512.02954) find from joint eROSITA X-ray gas fractions plus SDSS/DESI+ACT kSZ profiles a matter power suppression of 10 +/- 2% at k = 1 h/Mpc, stronger than the fiducial FLAMINGO and BAHAMAS models. HSC-XXL gas fractions prefer 5 +/- 4% at k = 1 h/Mpc.	tc-261023_1-w1-d-10-42

Id	Confidence	Claim	Evidence (tool calls, figures)
f-021	high, **contested**	Grandis et al. (arXiv:2309.02920) use cluster gas and stellar fractions to infer a suppression of 0.042 (+0.012/-0.014) at k = 1 h/Mpc and 0.184 (+0.026/-0.031) at k = 3 h/Mpc. They predict, at 97.5% credibility, that baryons change P(k) by less than 1% for k < 0.37 h/Mpc, and on that basis question whether baryon feedback drives the cosmic-shear vs CMB S8 discrepancy.	tc-261023_1-w1-d-10-36
f-022	high, **contested**	Preston, Amon and Efstathiou (arXiv:2305.09827) model P_m = P_L + A_mod (P_NL - P_L). Their KiDS-1000 Paper I value is A_mod ≈ 0.69. For DES Y3 the abstract says only that the required suppression is "not as strong as inferred using KiDS data, but is still more extreme than predictions from recent numerical simulations", and that it must extend to k ~ 0.2 h/Mpc. The DES Y3 numerical A_mod was not in the portion of the text read so far.	tc-261023_1-w1-d-10-41
f-023	high	Maraio, Hall and Taylor (arXiv:2410.12500) find that for a Euclid-like survey with ell_max = 5000, even multi-parameter baryon models (including HMCode-2020) can bias Omega_m and S8 significantly. A theoretical-error covariance on P(k) reduces this bias to acceptable levels with only a modest increase in parameter variances. The abstract gives no numerical S8 shift.	tc-261023_1-w1-d-10-43
f-024	high, **contested**	At z=0 the combined extreme HACC point (kappa_w=2, e_w=1.0, M_seed=2.0, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) gives P_hydro/P_grav = 0.99944+/-0.00003 at k=0.101, 0.9868+/-0.0008 at k=0.510, 0.9638+/-0.0020 at k=1.013, 0.9179+/-0.0045 at k=2.002, 0.8698+/-0.0065 at k=3.007 and 0.8400+/-0.0076 at k=3.698 h/Mpc. That is a suppression of 3.6+/-0.2% at k=1 and 13.0+/-0.7% at k=3. Against HSC-XXL (5+/-4% at k=1) the gap is 0.35 sigma, so the point is consistent. Against eROSITA+kSZ (10+/-2% at k=1) it falls short by 3.2 sigma. Against 2309.02920 (4.2% at k=1, 18.4% at k=3) it is 0.6 po	tc-261023_1-w1-d-08-29, tc-261023_1-w1-d-08-45, tc-261023_1-w1-d-08-46

Id	Confidence	Claim	Evidence (tool calls, figures)
f-025	high	Answers q-07. Preston, Amon and Efstathiou (arXiv:2305.09827, Eq. 3) find that Planck LCDM is compatible with DES Y3 cosmic shear if non-linear power is suppressed with A_mod = 0.858 +/- 0.052 (no scale cuts, Planck prior on S8 and Omega_m). This is a weaker suppression than the KiDS-1000 value A_mod ≈ 0.69 from Paper I (arXiv:2206.11794).	tc-261023_1-w1-d-10-57, tc-261023_1-w1-d-10-41
f-026	medium	fGas (cluster gas fraction vs M500c, 3 mass bins) was predicted at z=0 for the fiducial, strongAGN and weakAGN points. The CSVs are written, but their values were not read because the tool budget ran out, so the van Daalen-type pairing of gas fraction with suppression is still open.	tc-261023_1-w1-d-09-61, tc-261023_1-w1-d-09-62, tc-261023_1-w1-d-09-63
f-027	medium	For the combined extreme HACC point, suppression weakens strongly from z=0 to z=1. At k=1.013 h/Mpc the ratio rises from 0.9638+/-0.0020 at z=0 to 0.9781+/-0.0018 at z=1, so the suppression drops from 3.6% to 2.2%. At k=3.007 it rises from 0.8698+/-0.0065 to 0.9299+/-0.0055 (13.0% to 7.0%). At k=3.70 it rises from 0.8400+/-0.0076 to 0.9214+/-0.0064 (16.0% to 7.9%). Within k of 3.7 h/Mpc the response therefore grows as z falls, which agrees qualitatively with FLAMINGO (2410.17109). The ratio is still falling at k=3.7 at both redshifts, so the minimum lies beyond the range read here and its shif	tc-261023_1-w1-d-08-45, tc-261023_1-w1-d-08-60, tc-261023_1-w1-d-08-47 Fig. 28
f-028	high	Four validated cite entries (arxiv230509827, arxiv241012500, arxiv251202954, arxiv230902920) were written to /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/references.bib, each with a verbatim abstract quote.	tc-261023_1-w1-d-10-65, tc-261023_1-w1-d-10-69, tc-261023_1-w1-d-10-70, tc-261023_1-w1-d-10-71

Id	Confidence	Claim	Evidence (tool calls, figures)
f-029	medium, **contested**	Corrected deficit significance for the HACC extreme point (f-024), with emulator and published errors combined in quadrature and the published error taken on the side facing the HACC value. Against Grandis et al. (2309.02920): at k=1, (4.2-3.6)/sqrt(0.2^2+1.4^2) = 0.4 sigma; at k=3, (18.4-13.0)/sqrt(0.7^2+3.1^2) = 1.7 sigma. Against eROSITA+kSZ (2512.02954): (10-3.6)/sqrt(0.2^2+2^2) = 3.2 sigma. Against HSC-XXL: 1.4/sqrt(0.2^2+4^2) = 0.35 sigma. This replaces the f-024 statements of about 3 sigma and 8 sigma, which counted emulator error only.	tc-261023_1-w1-d-08-29, tc-261023_1-w1-d-08-45, tc-261023_1-w1-d-10-36, tc-261023_1-w1-d-10-42
f-030	high	Full texts of FLAMINGO (2410.17109), Siegel et al. (2512.02954) and Grandis et al. (2309.02920) were downloaded to /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-baryonic-pk/runs/261023_143127/2410.17109.txt, 2512.02954.txt and 2309.02920.txt. They have not been read yet, so no verbatim quote has been checked and no FLAMINGO S(k) value is sourced.	tc-261023_1-w2-d-12-81, tc-261023_1-w2-d-12-82, tc-261023_1-w2-d-12-85
f-031	high, **contested**	Maraio, Hall and Taylor (arXiv:2410.12500, Table 3) work with an idealised Euclid-like (Stage-IV) cosmic-shear survey at ell_max = 5000 and use Illustris as the ground truth. When the one-parameter HMCode baryon model is fitted without a theory error, Omega_m is biased by 100*Delta Omega_m = -1.317, which is -5.690 sigma. Adding the theoretical-error covariance cuts this to 0.128 (0.249 sigma). With Bahamas as the ground truth the bias is only 0.057 (0.258 sigma). Eagle with the 6-parameter model is still biased with the theory error included: 0.573 (1.459 sigma). In the text, the one-paramete	tc-261023_1-w2-d-13-86, tc-261023_1-w2-d-13-75

Id	Confidence	Claim	Evidence (tool calls, figures)
f-032	high	For the combined extreme point, the HACC maxsupp z=0 Pk-ratio keeps falling past k=3.7 h/Mpc: 0.8284+/-0.0079 at k=4.00, 0.7942+/-0.0089 at k=5.00, 0.7680+/-0.0095 at k=6.0, 0.7479+/-0.0100 at k=7.01 and 0.7356 at k=7.84 h/Mpc. That is already deeper than the strongAGN plateau of about 0.789 at k of about 12 h/Mpc (f-017), so kappa_w=2 with M_seed=2.0 gives more small-scale suppression than kappa_w=3. The values at k=0.2-0.3 h/Mpc were not read within the budget.	tc-261023_1-w2-d-13-83
f-033	high	The HACC fGas emulator at z=0 has 3 M500c bins: 5.01e13, 7.94e13 and 1.26e14 Msun/h. In the bin nearest 1e14 (1.26e14 Msun/h), f_gas = 0.1214+/-0.0010 for fiducial, 0.1221+/-0.0019 for strongAGN and 0.1414+/-0.0023 for weakAGN (e_w=0.2, v_kin=0.1, eps_kin=0.02). At 7.94e13 Msun/h the values are 0.1090, 0.1098 and 0.1375.	tc-261023_1-w2-d-11-76, tc-261023_1-w2-d-11-78, tc-261023_1-w2-d-11-79
f-034	medium, **contested**	Partial answer to q-12. Only two fGas points have S(k) read so far. Fiducial: f_gas(1.26e14) = 0.1214+/-0.0010, with S(k=1) = 0.9779 and S(k=3) = 0.9197. StrongAGN: f_gas = 0.1221+/-0.0019, with S(k=1) = 0.9746 and S(k=3) = 0.9121. The two f_gas values agree within errors; strongAGN is nominally higher, yet its suppression is 0.3-0.8 points deeper. Between these two points there is therefore no resolvable van Daalen-type relation, and the nominal trend runs the wrong way. WeakAGN has a much higher f_gas (0.1414), but its S(k) was not read in this direction, so a three-point test is still open.	tc-261023_1-w2-d-11-76, tc-261023_1-w2-d-11-78, tc-261023_1-w2-d-11-79, tc-261023_1-w1-d-09-38, tc-261023_1-w1-d-09-25
f-035	high	The HACC maxsupp_lowkin point at z=0 (kappa_w=2, e_w=1.0, M_seed=2.0, v_kin=0.1, eps_kin=0.02, omega_m=0.14, sigma_8=0.8) shows essentially no suppression. P_hydro/P_grav is 1.0002 at k=0.212, 1.0004+/-0.0007 at k=0.510, 1.0006+/-0.0019 at k=1.013, 1.0033+/-0.0042 at k=2.002, 0.9972+/-0.0062 at k=3.007 and 0.9883+/-0.0071 at k=3.698 h/Mpc. So in this emulator the kinetic AGN parameters (v_kin, eps_kin) drive almost all of the suppression at k up to 3.7 h/Mpc, even with e_w at its maximum.	tc-261023_1-w2-d-11-84, tc-261023_1-w2-d-11-87

Id	Confidence	Claim	Evidence (tool calls, figures)
f-036	high	The HACC maxsupp_lowM point at z=0 (kappa_w=2, e_w=1.0, M_seed=0.6, v_kin=1.2, eps_kin=1.2, omega_m=0.14, sigma_8=0.8) suppresses more than the combined maxsupp point (M_seed=2.0). P_hydro/P_grav is 0.9967 at k=0.212, 0.9835+/-0.0011 at k=0.510, 0.9557+/-0.0029 at k=1.013, 0.9026+/-0.0064 at k=2.002, 0.8521+/-0.0094 at k=3.007 and 0.8228+/-0.0109 at k=3.698 h/Mpc. The maxsupp values are 0.9638, 0.8698 and 0.8400 at k=1, 3 and 3.7. The suppression is 4.4+/-0.3% at k=1 and 14.8+/-0.9% at k=3, so maxsupp is not the strongest point, and low M_seed increases the suppression. This is still short of	tc-261023_1-w2-d-11-88, tc-261023_1-w1-d-08-29, tc-261023_1-w1-d-10-42

B.3 Dead ends

- x-01 (d-05): Reading the HACC Pk-ratio CSVs beyond k of about 3.73 h/Mpc with read_text_file, in order to get the ratio at k=5 and 10 h/Mpc and the suppression minimum. — read_text_file has no offset parameter and truncates each 799-bin CSV near k=3.73 h/Mpc. In all of the new runs (kappa_w=2, kappa_w=4, M_seed=2) the suppression is still deepening monotonically at that k. The minimum and any high-k upturn therefore remain unresolved above k=3.73 h/Mpc. A header-free copy for load_table_data would need a file-writing tool, which is not available. The comparison plo

- x-02 (d-10): get_paper_figures on arXiv:2410.17109 (FLAMINGO baryon effects) for Figs. 2, 5, 6 to read the depth of the z=0 minimum. — Neither the arXiv HTML nor the ar5iv rendering has extractable figures. Reading the depth would need the PDF and text, so q-08 stays open.

- x-03 (d-08): Making two separate comparison plots (the z=0 parameter variants, and z=0 versus z=1) with plot_observable_comparison and the same output_dir. — Both calls wrote the same file, Pk-ratio_comparison.png, and both were registered as Figure 28. The second call (z=0 versus z=1 for fiducial and maxsupp) overwrote the first (fiducial, maxsupp, maxsupp_lowkin and maxsupp_lowM at z=0). The tool has no filename or label argument, so each plot needs its own output_dir subdirectory.

- x-04 (d-09): Loading the HACC Pk-ratio prediction CSVs with load_table_data — The CSV header row 'x,y_mean,y_std' after the '#' comments is not skipped, and the tool fails with "could not convert string 'x' to float64". It has no skiprows option. Workaround: read_arxiv_chunk with start and length offsets reads the whole file (about 40 kB, up to k=12.6 h/Mpc).

B.4 Critic verdicts

- c-01 (wave 0, plan): **revise**; medium: The HACC Pk-ratio output may not cover k=0.1 to 10 h/Mpc, and its redshifts may be a discrete list. After list_observables or describe_parameters, record the actual k range and report the minimum only if it lies inside

that range. Do not extrapolate. If the curve has no turnover before k_max, state ; medium: Robustness: at k=1 h/Mpc and at the minimum, compare the difference between the strong-feedback and weak-feedback curves with the emulator uncertainty band. State whether the variants can be distinguished.; medium: Interpretation guard: HACC subgrid parameters are not the same as the FLAMINGO fgas or jet variants, nor the BCemu log10 Mc. Present the comparison with the 5 +/- 4 percent and 10 +/- 2 percent constraints (2512.02954) as a range-of-model check, not a calibration. Do not attribute any computed numbe; high: Over budget. Five or six papers, each needing download_arxiv_full_text, several read_arxiv_chunk calls and cite, plus get_paper_figures, comes to well over 20 calls. Narrow d-02 to the AGN feedback physics: 2410.17109 (full text, one figure), 2309.02920 or 2309.06479 for halo mass and baryon fractio; medium: No tool here computes C_ell, xi_plus or an S8 forecast. Any S8 bias or ell-range statement must be a sourced quote, labelled as literature. The finding must not present it as a computed result.; high: Over budget. Roughly 2 to 3 AGN parameters x 2 values, plus 2 cosmology runs, 2 to 3 redshifts, 3 fGas calls, 2 plots and describe_parameters comes to about 17 or more calls. Trim to: the single most influential AGN parameter at low and high (2 calls, reusing the d-01 runs where possible), one cosmo

- `c-02` (wave 1, wave): **continue**; contested `f-002`, `f-003`; medium: The quoted CSV excerpts stop at k=0.117 h/Mpc. The values at k=0.51, 1, 2, 3 and 3.71 are not visible in the cited evidence; only the k=0.101 values (0.99968 fiducial, 0.99962 strong) can be checked. The k=1 values are the key numbers of the task and should be re-extracted with a working reader and ; medium: Wording of the comparison. The phrase 'eROSITA+kSZ ... excluded at more than 3.5 sigma' reverses the logic. The emulator cannot exclude a measurement. The correct statement is that the sampled HACC points fall short of the eROSITA+kSZ value by about 3.7 sigma (observational error only). The comparis; high: There is a physics discrepancy no one has noted. At k=3 h/Mpc the HACC fiducial and strong points give about 8-9 percent suppression, while the cluster-calibrated prior of 2309.02920 gives 18.4 percent (+2.6/-3.1). At k=1 the emulator gives 2.2-2.5 percent against 4.2 percent there and 5-10 percent ; medium: Two sourcing gaps remain (q-05). 2309.02920 has no stored citation or quote. The depth and k of FLAMINGO's z=0 minimum, which the brief listed as a quantity to check, is still unsourced, and so are the 1104.1174 claims of a 10-30 percent minimum at 5-20 h/Mpc and the high-k upturn.; low: 'Consistent with Planck LCDM only if' is stronger than the source. Amon and Efstathiou show that suppression with A_mod about 0.69 is sufficient, not that it is the only route. 'Costs 2x in precision' should be stated precisely: the S8 uncertainty roughly doubles.; low: f-005 was written before the d-01 numbers were available and is now stale. It should be replaced by an explicit comparison table: HACC at k=1 and k=3 versus 2309.02920, HSC-XXL and eROSITA+kSZ.

- `c-03` (wave 2, wave): **continue**; contested `f-007`, `f-008`, `f-011`; low: The quoted phrases 'not as strong as inferred using KiDS data' and 'k ~ 0.2 h/Mpc' are attributed to the abstract, but the excerpt from tc-...-72 is cut off after the author affiliations. Only the conclusions chunk (tc-...-78) visibly supports 'stronger than favoured in recent hydrodynamic simulatio; medium: The visible excerpts support the fgas range from -10 to +4 sigma and the statement that only Illustris and cosmo-OWLS 8.7 are out of reach. The excerpts do not show the minimum positions (k=6 at z=2, k=10 at z=0), the saturation below fgas-8sigma, the Bigwood et al. compatibility, or the claim that ; medium: The excerpts for every CSV are cut off at k of about 0.12 h/Mpc, so the k=1 and k=3 values cannot be checked against the quoted evidence. They are internally consistent with each other and with f-001. e_w, v_kin and eps_kin were only moved jointly, so their individual sensitivities are unknown. omeg; high: The task asks for the shape of the suppression up to about 10

h/Mpc, including the depth and position of the minimum, and this is still not delivered. All runs are still deepening at k=3.7 h/Mpc because of the CSV truncation (x-01, q-06). Without the curve above 3.7 h/Mpc, the comparison of minimum ; medium: f-002 is still contested on the board. Its claim that the k=1 suppression spans -0.2% to +2.5% across the full AGN design range is falsified by kappa_w=2 (2.92%). It should be formally retired in favour of f-015.; medium: The weak-lensing assessment has no projected quantity. There is no C_ell or xi_+ suppression and no S8 shift. The only ell range comes from an unverified f-007 statement (ell_max = 5000). The lensing-weighted redshift of about 0.3-0.5 was never probed, since all HACC runs are at z=0.

- `c-04` (wave 2, report): **revise**; high: unsupported: The wind-coupling parameter kappa_w has the largest effect at k=3. — This contradicts f-014, and the report's own ranking. At k=3 the joint lowering of e_w, v_kin and eps_kin (about -8.8 pp) is larger than the kappa_w spread (5.98 pp). kappa_w is the largest single-parameter one-at-a-ti; high: unsupported: Even the strong-feedback eROSITA+kSZ result (10 +/- 2% at k=1) falls short of what the FLAMINGO emulator can produce for the Amon–Efstathiou response (arXiv:2410.17109). — f-010 states only that the Amon-Efstathiou response is out of reach of the FLAMINGO emulator. The board has no comp; high: unsupported: about 0.03–0.04% suppression at k=0.1 h/Mpc; about 0.8–0.9% at k=0.5 h/Mpc — This is presented as holding within the whole sampled range. f-008 derives these values from the fiducial and strong points only. The weak point shows an enhancement at these k. No k=0.1 or k=0.5 values for the; high: unsupported: This is consistent with the literature picture in which the concentration of baryons in halo centres dominates once ejective feedback is weak (arXiv:2309.02920). — The board sources 2309.02920 only for star formation concentrating baryons on small scales. Using it to explain an emulator; high: unsupported: Suppression appears at about k?0.3 h/Mpc for the fiducial and strong points. The two curves nearly coincide, and the weak-AGN curve sits slightly above unity. — No board entry describes the content of Figure 8 or an onset scale of k~0.3 h/Mpc. The tabulated values give only k=0.1 and k=

- `c-05` (wave 0, plan): **revise**; high: Three corner-style points cannot establish the emulator's maximum suppression. Report the result as 'largest suppression among the sampled points'. Alternatively, spend the spare calls on a few extra corners, for example with kappa_w and e_w at their bounds and the kinetic parameters varied, before ; high: Step 1 must confirm, before any run, that every value lies inside the describe_parameters bounds. This applies to e_w=1.0, v_kin=1.2, eps_kin=1.2, M_seed=2.0 and the cosmology omega_m=0.14, sigma_8=0.8. Points at the corners of the design hypercube carry the largest emulator error, so the 1-sigma mu; medium: Confirm the emulator's k units (h/Mpc or 1/Mpc) before comparing with HSC-XXL, eROSITA+kSZ and 2309.02920. Sigma comparisons should combine the emulator 1-sigma and the observational error in quadrature, and the method should be stated. The HSC-XXL and eROSITA+kSZ numbers (5+/-4% and 10+/-2%) curren; low: Check that predict_observable supports z=1 for Pk-ratio. If only a nearby redshift is available, label it with its exact value. FLAMINGO's redshift trend should be compared only qualitatively, because the parameterisations differ.; high: The cutoff at k=3.7 h/Mpc may be the emulator's training k_max rather than a truncation by the text reader. Check describe_parameters or list_observables for the Pk-ratio k range first. If k_max is about 3.7, record that the high-k minimum and the upturn are outside the emulator's validity, and skip; medium: Confirm the fGas mass-bin range and units (Msun versus Msun/h, and M500 versus M200) before reading off values at about 1e14. A monotonicity claim based on four points is qualitative only. Use the same four parameter points as the Pk-ratio runs so the pairing is like-for-like. The call budget is tig

- `c-06` (wave 1, wave): **continue**; contested `f-024`, `f-019`, `f-020`, `f-021`, `f-022`; high: The

maximum suppression is not established. The maxsupp_lowkin and maxsupp_lowM CSVs exist but were never read. Cosmology was held at omega_m=0.14 and sigma_8=0.8, although the suppression depends on the cosmic baryon fraction. The kappa_w effect is confounded with M_seed: maxsupp (kappa_w=2, M_seed; high: The fGas CSVs were written but never read. The pairing of gas fraction with suppression is the central quantitative test of the AGN-physics explanation the task asks for, and reading the files costs about 3 tool calls.; medium: The z=0 to z=1 trend has been measured for one point only; fiducial_z1 was not read. Two FLAMINGO statements are unsourced while q-08/q-13 stays open: the 'agrees with FLAMINGO' claim, and the statement that FLAMINGO's minimum lies at k~8-10 h/Mpc.; medium: q-09 is unresolved. The P_grav reference used by HACC may differ from the P_DMO reference used in the literature, and this should be checked before the deficit at k=3 is interpreted.; medium: The weak-lensing assessment is still qualitative. No numerical S8 shift has been extracted. The HACC suppression at k=0.2-0.5 has not been compared with the Preston et al. requirement that suppression extend to k~0.2 h/Mpc.; low: The k_max of about 12.6 h/Mpc and the plateau at about 0.789 are inferred from byte offsets; the quoted excerpt stops at k=9.70. Quote the final rows of the file verbatim.

- `c-07` (wave 2, wave): **continue**; contested `f-029`, `f-031`, `f-034`; high: Every literature-comparison significance on the board (f-024, f-029, f-036) depends on f-020 and f-021, and neither has been checked verbatim. The full texts are already on disk, so checking them is cheap. Until it is done, the headline conclusion that HACC falls short of eROSITA+kSZ by about 3 sigm; medium: The emulator values for lowM are supported. The comparisons to Siegel et al. and Grandis et al. inherit the contested status of f-020 and f-021. The claim that 'the design range does not reach eROSITA+kSZ' has only been tested at the kappa_w=2, e_w=1 corners with omega_m=0.14 and sigma_8=0.8. The lo; medium: There is a physically interesting signal here that the finding does not develop. Strong kinetic feedback leaves f_gas(<R500c) at 1.26e14 essentially unchanged (0.1214 vs 0.1221), yet it deepens S(k). That suggests the HACC suppression is driven by gas removal beyond R500c, or by other halo masses, r; low: Attributing the deeper suppression to 'kappa_w=2 vs kappa_w=3' is confounded, because M_seed also differs (2.0 vs 1.3). Note too that f-036 shows lower M_seed increases suppression. That makes the kappa_w attribution plausible, but it has not been isolated by changing one parameter at a time.; medium: The weak-lensing assessment is still qualitative. There is no Delta S8 value (Table 4 of 2410.12500 is unread), the HACC k=0.2-0.3 values are interpolated rather than read, and there is no A_mod-equivalent number for any HACC point. Two directions in a row hit their budget limit before reading the f
- `c-08` (wave 2, report): **revise**; high: unsupported: Figure 28 shows the emulator Pk-ratio at z = 0 and z = 1 for the fiducial and maxsupp points. For both points the z = 1 curves lie closer to unity than the z = 0 curves. — The board says the fiducial z=1 values were never read (d-08, q-16b), and it gives no description of what Figure 28; high: unsupported: fiducial (3, 0.6, 1.3, 0.65, 0.61) — The board defines the fiducial point only as kappa_w=3, e_w=0.6, v_kin=0.65, eps_kin=0.61 (f-019). No M_seed value for fiducial is recorded, so M_seed=1.3 is not sourced.; high: unsupported: All points used omega_m = 0.14 and sigma_8 = 0.8. — The board lists this cosmology for maxsupp, maxsupp_lowkin, maxsupp_lowM and strongAGN. The fiducial and weakAGN cosmologies are not stated (f-019, f-033).; high: unsupported: Against observations, the sampled HACC points sit with the "standard" feedback — The report is truncated at this point. The Discussion is unfinished and there is no conclusion or open-directions section. The incomplete characterisation of 'standard' feedback is also not stated on the bo; medium: missing finding f-028; medium: missing finding f-030